ST. LUCY DIGITAL PATIENT MANAGEMENT SYSTEM

GROUP 17

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PROJECT TITLE

Digital Patient Management System

BUSINESS CONTEXT

St. Lucy Medical Clinic is a health services provider situated at Kikumi Kikumi, Makerere, Uganda. It offers general medicine, dental services, minor surgeries, HIV counselling to mention but a few.

PROBLEM STATEMENT

The clinic currently operates using largely manual and paper-based processes across its key functional areas, including patient record management, out-patient appointment scheduling, financial transaction monitoring, and patient health education. These methods have increasingly proven to be inadequate in meeting the growing demands of the clinic's patient population and administrative workload.

PROJECT OBJECTIVES

1. To digitize and centralize 100% of patient medical records within 6 months, reducing record retrieval from over 10 minutes to less than 2 minutes, thereby improving efficiency and continuity of patience care.
2. To reduce average outpatient waiting time by 40% within 4 months by improving and streamlining the appointment scheduling process.
3. To improve financial accountability by implementing a system that enables real time tracking of 100% of clinic transactions within 5 months, reducing financial discrepancies by at least 30%.
4. To increase patient health awareness by ensuring that at least 60% of the registered patients access the e-learning platform within 6 months of deployment.

SYSTEM REQUIREMENTS SPECIFICATIONS

i) FUNCTIONAL REQUIREMENTS

Functional requirements define the core capabilities, operations, and specific tasks that the system must perform to support clinic workflows.

User Authentication

- The system shall allow users to register and login.

Appointment Management

- The system shall allow receptionists to schedule, update, and cancel appointments.

Patient Records Management

- The system shall allow doctors to view and update patient records.
- The system shall provide search functionality for patient records.
- The system shall store patient medical history.

Financial Management

- The system shall manage billing and payments.

Reporting

- The system shall generate reports for appointments and financial records.

Educational Integration

- The system shall allow patients to access e-learning platforms.

ii) NON-FUNCTIONAL REQUIREMENTS

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CONCEPTUAL DATABASE DESIGN

i) ENTITIES

An entity is the core object about which information is stored. It becomes a database table.

Entities of the Digital Patient Management System

User: Represents people who have authorized access to the system and whose accounts are managed by the administrator.

Patient: Represents patients registered and maintained in the clinical system.

Doctor: Represents doctors involved in appointments and clinical care.

Appointment: Represents a scheduled clinical appointment between a patient and doctor.

Medical Record: Represents the detailed clinical information maintained for a patient, including history, diagnosis and prescriptions.

Payment: Represents financial transactions made by patients for services rendered.

Attributes for each entity

Attribute is a specific characteristic or detail that describes that entity. It becomes a column in the table.

User

Attributes of user

- User ID
- Username
- Password
- Role

Patient

Attributes of Patient

- Patient ID
- Name
- Date Of Birth
- Gender
- Phone Number

Doctor

Attributes of Doctor

- Doctor ID
- Name
- Role

Appointment

Attributes of appointment

- Appointment ID
- Patient ID
- Doctor ID
- Appointment Date
- Appointment Time
- AppointmentStatus

Medical Record

Attributes of medical record

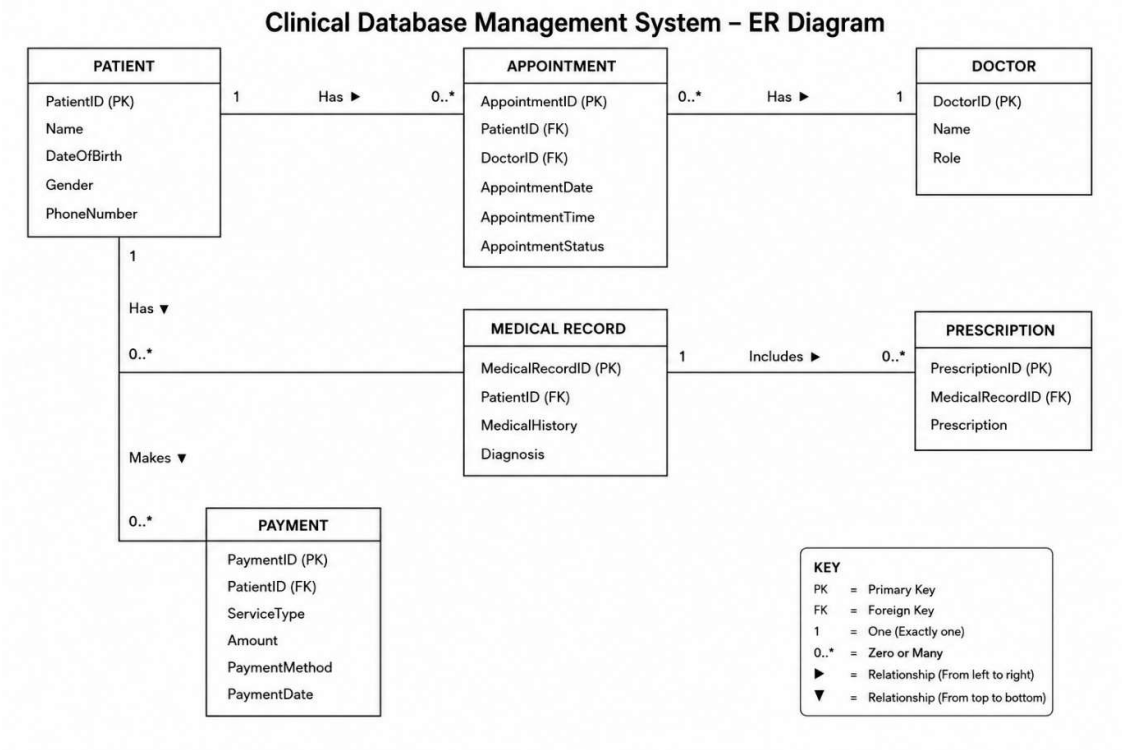
- Medical Record ID
- Patient ID
- Medical History
- Diagnosis
- Prescription

Payment

Attributes of Payment

- Payment ID
- Patient ID
- Service Type
- Amount
- Payment Method

ii) ENTITY RELATIONSHIP DIAGRAM



LOGICAL DATABASE DESIGN

1. Primary Key (PK)

A Primary Key is a column or set of columns that uniquely identifies each row in a table. It enforces entity integrity by ensuring that duplicate records cannot exist and that the key value can never be null.

- **Database Implementation:** Fields like patient_id or user_id serve as primary keys to enforce unique constraints in the PostgreSQL back-end.
- **Project Example (Patient Table):**
  - ✓ patient_id (Primary Key) for example, 1001, 1002, 1003 (Unique identifier).
  - ✓ name, date_of_birth, gender, phone_number.

2. Foreign Key (FK)

A Foreign Key is a column that establishes a link between data in two tables. It creates a relationship by pointing from a child table to the Primary Key of a parent table, ensuring data consistency across dashboards.

- **Database Implementation:** Enforces referential integrity, preventing actions that would destroy links between tables (For example, preventing an appointment from being booked for a non-existent patient).
- **Project Example (Appointment Table):**
  - ✓ appointment_id (Primary Key)
  - ✓ patient_id (Foreign Key) → References Patient(patient_id) to link a patient to a slot.
  - ✓ doctor_id (Foreign Key) → References Doctor(doctor_id) to assign a specific doctor.

Master Schema Relationship Matrix

Table Name	Primary Key (PK)	Foreign Keys (FK)	Relational Links (References)	Purpose in St. Lucy System
User	user_id	*None*	*None (Parent Table)*	Manages system access, login security, and user permissions.
Patient	patient_id	*None*	*None (Parent Table)*	Stores demographic and identity profiles for clinic patients.
Doctor	doctor_id	*None*	*None (Parent Table)*	Maintains records of available clinical staff for scheduling.
Appointment	appointment_id	patient_id, doctor_id	Patient(patient_id), Doctor(doctor_id)	Schedules time slots, enforcing links to existing patients and doctors.
Medical Record	medical_record_id	patient_id	Patient(patient_id)	Archives historical and active medical cases linked to a unique patient.
Payment	payment_id	patient_id	Patient(patient_id)	Manages revenue collection and billing details per patient.